Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Computational physicist (Ph.D.) with 10+ years writing scientific code, running computational experiments, and validating models against data. Currently evaluates frontier language models on expert-level physics reasoning and builds neural-network surrogates of expensive simulations. Author of open-source scientific libraries (Paramo, Tleco; ApJ 2024) with code review and unit testing, and mentor of graduate students.

Skills

Languages: Python, C/C++, R, Rust, SQL, Fortran 95, Java, Julia, Shell
ML & Data: Scientific machine learning, language-model evaluation, surrogate modeling, Bayesian inference, Gaussian process regression, PyTorch, scikit-learn, TensorFlow, random forest, neural networks, SVM, gradient-based optimization, ML-assisted parameter estimation, NumPy, SciPy, Pandas, Astropy
Geospatial: Sentinel-1/2 multispectral processing, DEM terrain analysis, raster/vector data pipelines, QGIS
HPC & Parallel: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5
Tools: Git, Docker, Azure, Jenkins, OpenShift, Kafka, Splunk, Jira, ParaView, VisIt

Experience

Independent Contractor, *Mercor* Jan 2026 – Present
Physics subject-matter expert evaluating frontier language models for multiple leading AI labs.

- Review 50+ expert-level physics, hydrodynamics, astrophysics, and cosmology problems for leading AI labs; write original prompts that stump frontier models and diagnose failure modes in their reasoning.

Spatial Data Scientist, *TealWaters* May 2025 – Jan 2026
Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

Independent Researcher & Open-Source Developer, *Self-directed* Jan 2024 – May 2025
Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, a Rust/Python toolkit that solves the Fokker–Planck equation coupled to radiative transfer to model radiation from relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.

Software Engineer, *Paychex* Apr 2022 – Jan 2024
Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

Postdoctoral Research Associate, *Rochester Institute of Technology* Feb 2021 – Apr 2022
HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries; added black hole spin with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo against Fermi-LAT gamma-ray spectra with gradient-based optimization that updates the model automatically; supervised a graduate student on the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron star mergers.

Postdoctoral Research Fellow, *Purdue University* Oct 2018 – Nov 2020
Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation for relativistic particle distributions coupled to radiative transfer; optimized it with OpenMP and tested GPU offload with OpenACC.
- Implemented implicit PDE solvers, numerical integration, and arbitrary-shape particle distributions without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization; other groups have adopted Paramo, including Combi & Siegel (2023).

- Served as co-I on two NASA Fermi Guest Investigator grant proposals, one funded ($68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

Postdoctoral Research Fellow, *Universidad Michoacana de San Nicolás de Hidalgo* Jan 2018 – Sep 2018

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Graduate Research Assistant, *Universitat de València* Oct 2011 – Jul 2017

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

Education

Universitat de València, *Ph.D. in Physics, Excellent Cum Laude* Jul 2017

Universidad Michoacana de San Nicolás de Hidalgo, *M.Sc. in Physics* Sep 2011

Universidad Autónoma del Estado de México, *B.Sc. in Physics* Dec 2008

Projects

Blazar Population ML Jan 2026 – Present

Scientific ML (Python, scikit-learn, PyTorch) github.com/altjerue/ilwikak-ml

- Applying classical ML and Gaussian process regression to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data.
- Building a neural-network emulator of the physics-based Tleco model that maps physical jet parameters to observable spectral energy distributions, to be validated against Tleco output (in progress).
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves (in progress).

Selected Publications (9 published articles, 3 first-author; 3 in preparation)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS 513, 5766–5779 (2022), arXiv:2201.07790.
- **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.
- Guzmán, F. S., & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D 80, 084023 (2009), arXiv:1009.1250.